README

DTMF Detection using Goertzel Algorithm
Overview
This project implements DTMF (Dual Tone Multi Frequency) detection using the Goertzel Algorithm in Python.
DTMF signals are used in telephone systems where each digit is represented by a combination of two frequencies.

Objective
To detect digits from a DTMF signal using an efficient frequency detection algorithm.

Methodology

- Generated DTMF tones using two sinusoidal frequencies
- Added silence between digits to simulate real signals
- Applied Goertzel algorithm to detect specific frequencies
- Used frame-based processing for detecting each digit
- Added noise and thresholding to improve robustness

Algorithm Used
Goertzel Algorithm
- Efficient method to detect specific frequencies
- Alternative to FFT when only few frequencies are required

How to Run

- You can run the code in online jupyter notebook

—

Output
The program detects the input number from the generated DTMF signal.

Example:
Detected Number: 08242474040

Files in Repository
- `dtmf_detection.py` → Main implementation code
- `README.md` → Project documentation

Author
Machavarapu Lahari

References
- Digital Signal Processing – Proakis & Manolakis
- DTMF Frequency Standards
- https://www.researchgate.net/publication/314804465_Dtmf_Detection_Using_Goertzel_Algorithm